

Laparoscopy is a minimally invasive surgical procedure where small surgical instruments are inserted into the body through cannulas (hollow tubes 0.5 to 1.0 cm in diameter). A cannula is inserted into the skin while sleeved around a trocar (Figure 1). The skin incision can be made directly with a sharp-tipped trocar as it is being inserted with the cannula; if a blunt-tipped trocar is used, the incision is made beforehand with a surgical blade. After insertion, the trocar is removed, leaving the cannula as an access point for laparoscopic procedures.

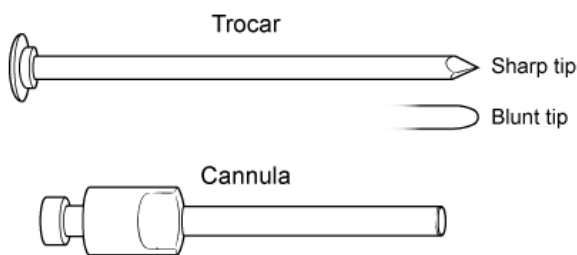


Figure 1 Cannula, trocar, and two types of tips

Low-friction trocars increase surgical precision and reduce postoperative scarring. Friction between the skin and trocar tip is tested with synthetic skin and a stationary ramp with a surface made of the same material as the trocar. For testing, the synthetic skin is pressed against the ramp's surface by a load, weighted to allow the skin to slide down at a constant speed (Figure 2). In this setup, the weight of the synthetic skin layer is negligible.

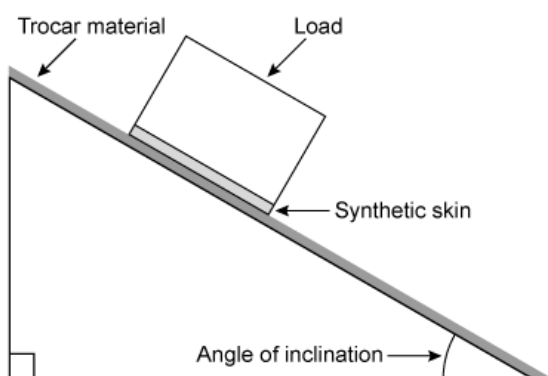


Figure 2 Experimental setup to test the friction between trocar and skin

For equally sized trocar-cannula systems, incisions made for blunt-tipped trocars are smaller than those made by sharp-tipped trocars, forcing the skin to stretch more around the cannula. This option better prevents slipping after the cannula is inserted because:

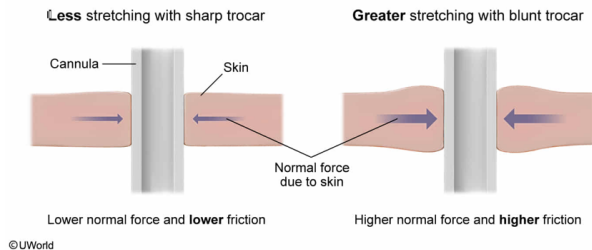
- ☐ A. it decreases the contact surface area. [12%]
- ☐ B. it decreases the static friction. [10%]

- ✓ ☒ C. it increases the normal force. [30%]
☐ D. it increases the coefficient of static friction. [46%]

Correct

31% answered correctly

Explanation:



Static friction is the frictional force that prevents two surfaces from sliding. After the cannula is inserted, **static friction** between the cannula and skin **prevents slipping** because static friction counters the forces that promote sliding. Static friction has an upper limit and if the forces that promote sliding exceed this value, slipping occurs. The **upper limit** of static friction F_s is the product of the coefficient of static friction μ_s and the normal force N (the perpendicular force one surface exerts on the other):

$$F_s = \mu_s N$$

The greater the upper limit of static friction, the better the cannula is prevented from slipping. Therefore, slipping can be better prevented by increasing the normal force. The normal force keeping the cannula in place is due to the skin pressing on the cannula. According to the question, cannulas inserted with blunt-tipped trocars experience greater stretching of the skin. Therefore, the cannula is subject to a **greater normal force**, producing a **greater static friction force**.

(Choice A) According to the question, the cannulas are of equal size, and therefore the sizes of the contact areas are the same.

(Choice B) Decreasing the static friction is more likely to cause the cannula to slip.

(Choice D) The coefficient of static friction depends only on the properties of the surfaces involved. Since both situations involve the same two surfaces (cannula and skin), the coefficient of static friction is the same.

Educational objective:

Friction is the force that resists sliding between two surfaces, and static friction prevents surfaces from sliding. The maximum value of static friction is proportional to the normal force, which is the perpendicular force one surface exerts on the other. Therefore, greater stretching of skin creates a greater normal force around a cannula

and greater static friction, which better prevents slipping.

Time spent:0 sec

Subject:
Physics

QID:401016

System:
Mechanics and Energy

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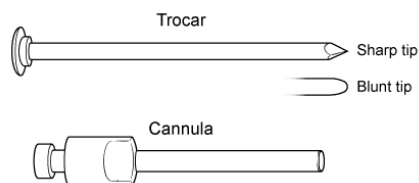


Figure 1 Cannula, trocar, and two types of tips

Low-friction trocars increase surgical precision and reduce postoperative scarring. Friction between the skin and trocar tip is tested with synthetic skin and a stationary ramp with a surface made of the same material as the trocar. For testing, the synthetic skin is pressed against the ramp's surface by a load, weighted to allow the skin to slide down at a constant speed (Figure 2). In this setup, the weight of the synthetic skin layer is negligible.

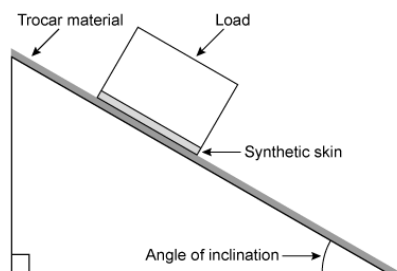
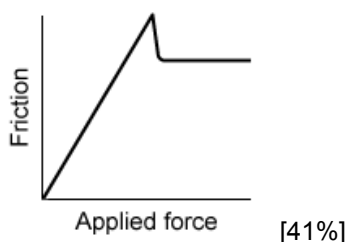


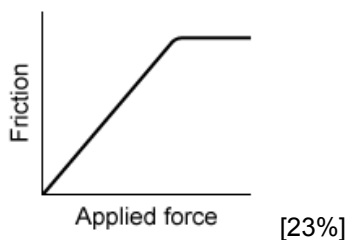
Figure 2 Experimental setup to test the friction between trocar and skin

To remove the cannula at the end of a laparoscopy, the surgeon pulls on the cannula with increasing force. How does the frictional force vary with the applied pulling force before and after the cannula begins to slide?

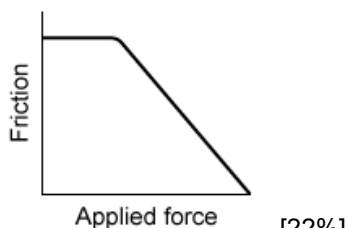
✓ ☒ A.



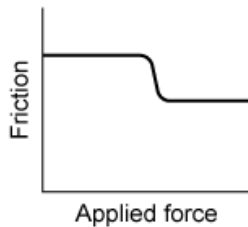
☐ B.



☐ C.



☐ D.



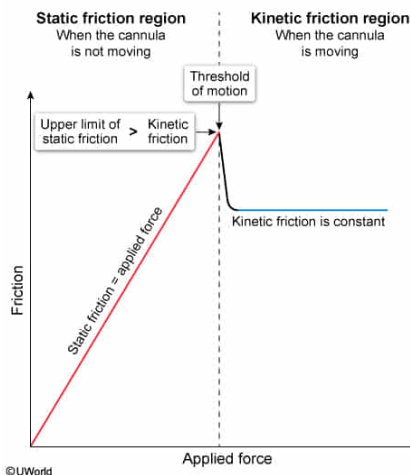
[4470]

[13%]

Correct

41% answered correctly

Explanation:



Friction is the force that resists sliding between two surfaces, and it can be static or kinetic. Static friction F_s occurs when surfaces do not slide because the magnitude of static friction fully counteracts an applied force (up to an upper limit). The upper limit of static friction is the product of the **normal force** N (the perpendicular force one surface exerts on the other) and the coefficient of static friction μ_s (a number unique to the two surfaces):

$$F_s = \mu_s N$$

When the applied pulling force first acts on the cannula, the cannula does not move due to static friction. During this time, the magnitude of static friction is equal to the magnitude of the applied force but less than its upper limit. As the applied force (x-axis) increases, **friction** (y-axis) **increases** until the **upper limit is reached**, which is the threshold for motion.

At this point, the cannula begins to slide, and static friction is replaced with kinetic friction. Kinetic friction occurs when surfaces slide against each other; it has a constant magnitude of:

$$F_k = \mu_k N$$

where μ_k is the coefficient of kinetic friction, which is **always less than μ_s** . When the cannula begins to slide, **friction** (y-axis) **suddenly drops** and **remains constant** (horizontal line) as the applied force increases.

(Choices B, C, and D) The graph should initially increase, as static friction is equal in magnitude to the applied force. When the upper limit of static friction is reached, the graph should drop to a constant kinetic friction.

Educational objective:

Static friction prevents an object from sliding and is equal in magnitude to the applied force until it reaches its upper limit ($F_s = \mu_s N$). When static friction is overcome, the frictional force is described by kinetic friction, which is constant and lower ($F_k = \mu_k N$). On a friction vs. applied force graph, static friction is shown as an increasing force that transitions into kinetic friction with a sudden drop to a constant value.

Time spent: 0 sec

Subject:
Physics

QID: 401017

System:
Mechanics and Energy

Laparoscopy is a minimally invasive surgical procedure where small surgical instruments are inserted into the body through cannulas (hollow tubes 0.5 to 1.0 cm in diameter). A cannula is inserted into the skin while sleeved around a trocar (Figure 1). The skin incision can be made directly with a sharp-tipped trocar as it is being inserted with the cannula; if a blunt-tipped trocar is used, the incision is made beforehand with a surgical blade. After insertion, the trocar is removed, leaving the cannula as an access point for laparoscopic procedures.

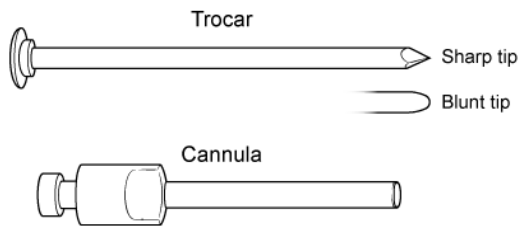


Figure 1 Cannula, trocar, and two types of tips

Low-friction trocars increase surgical precision and reduce postoperative scarring. Friction between the skin and trocar tip is tested with synthetic skin and a stationary ramp with a surface made of the same material as the trocar. For testing, the synthetic skin is pressed against the ramp's surface by a load, weighted to allow the skin to slide down at a constant speed (Figure 2). In this setup, the weight of the synthetic skin layer is negligible.

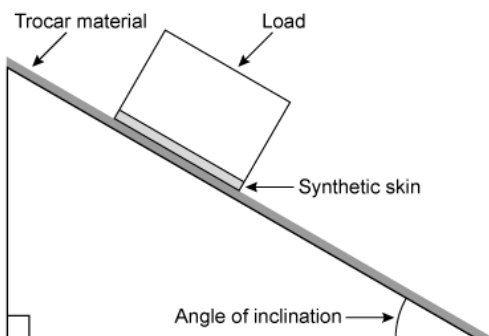


Figure 2 Experimental setup to test the friction between trocar and skin

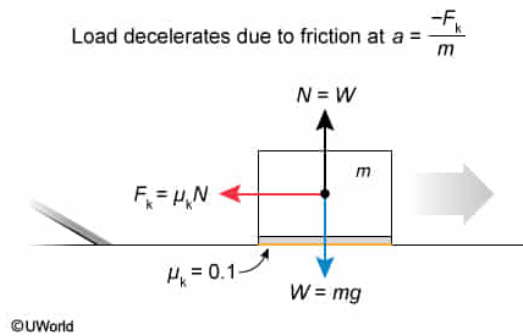
After the load leaves the ramp, it slides on a horizontal table with a coefficient of kinetic friction μ_k of 0.1 and a coefficient of static friction μ_s of 0.3. If the direction of the load's velocity is positive, what is the acceleration of the load? (Note: Use $g = 10 \text{ m/s}^2$.)

- ☐ A. -3 m/s^2 [10%]
- ✓ ☒ B. -1 m/s^2 [44%]
- ☐ C. 1 m/s^2 [28%]
- ☐ D. 3 m/s^2 [16%]

Correct

43% answered correctly

Explanation.



As the load slides on the table after leaving the ramp, friction resists the sliding motion.

Kinetic friction occurs when surfaces slide against each other, and it is the product of the **normal force** N (the perpendicular force one surface exerts on another) and the coefficient of kinetic friction μ_k (a dimensionless number unique to the two surfaces). On a horizontal surface, the normal force is equal to the magnitude of the object's **weight** ($N = W = mg$). Therefore, the magnitude of the **frictional force** acting on the sliding load is:

$$F_k = \mu_k N = \mu_k mg$$

According to **Newton's second law of motion**, an object experiences an **acceleration** a equal to the net force F_{net} divided by the object's mass m :

$$a = \frac{F_{\text{net}}}{m}$$

Because the direction of the load's velocity is positive and friction acts in the direction opposite the force, the net **force and acceleration are both negative** ($F_{\text{net}} = -F_k$). The acceleration of the load is calculated as the negative of the kinetic friction divided by the object's mass:

$$a = \frac{-\mu_k mg}{m} = -\mu_k g$$
$$a = -(0.1)(10 \text{ m/s}^2) = -1 \text{ m/s}^2$$

Therefore, the acceleration of the load is **-1 m/s²**.

(Choice A) The coefficient of static friction μ_s is used when surfaces are not sliding. Because the load is sliding, the coefficient of kinetic friction μ_k is used to calculate the frictional force acting on the load.

(Choices C and D) Because friction resists sliding, kinetic friction always decelerates sliding objects. Therefore, the acceleration of the load is negative, and the velocity of the object decreases over time.

Educational objective:

Kinetic friction occurs when two surfaces slide against each other; it is the product of the coefficient of kinetic friction and the normal force ($F_k = \mu_k N$). Because friction opposes sliding, kinetic friction causes sliding objects to decelerate. The magnitude of deceleration is calculated using Newton's second law of motion: $F = ma$.

Time spent:0 sec

Subject:
Physics

QID:401018

System:
Mechanics and Energy

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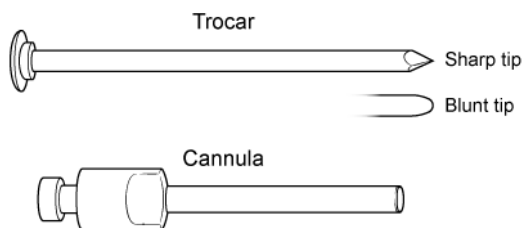


Figure 1 Cannula, trocar, and two types of tips

Low-friction trocars increase surgical precision and reduce postoperative scarring. Friction between the skin and trocar tip is tested with synthetic skin and a stationary ramp with a surface made of the same material as the trocar. For testing, the synthetic skin is pressed against the ramp's surface by a load, weighted to allow the skin to slide down at a constant speed (Figure 2). In this setup, the weight of the synthetic skin layer is negligible.

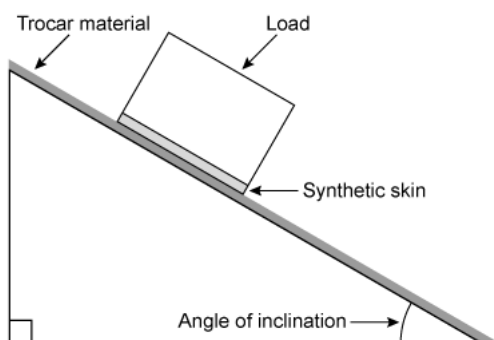


Figure 2 Experimental setup to test the friction between trocar and skin

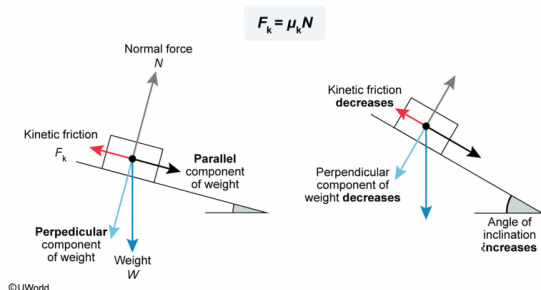
When the angle of inclination of the ramp increases, kinetic friction:

- ✔ ☒ A. decreases because the weight component perpendicular to the ramp decreases. [49%]
- ☐ B. decreases because the weight component parallel to the ramp decreases. [12%]
- ☐ C. increases because the weight component perpendicular to the ramp increases. [20%]
- ☐ D. increases because the weight component parallel to the ramp increases. [17%]

Correct

50% answered correctly

Explanation:



Kinetic friction F_k is the frictional force between two objects sliding against each other, and it is the product of the coefficient of kinetic friction μ_k (specific to the two surfaces) and the **normal force** N (the perpendicular force one surface exerts on another):

$$F_k = \mu_k N$$

When the mass slides down a ramp with angle of inclination θ , its **weight** W can be split into two **component force vectors**, the sum of which is the weight vector. One component of the weight is parallel to the ramp's surface ($W \sin \theta$) and points down the ramp, causing the mass to slide. The other component is perpendicular to the ramp's surface ($W \cos \theta$) and keeps the mass in contact with the ramp. The normal force is equal and opposite to this perpendicular component:

$$N = W \cos \theta$$

Substituting this expression into the above equation for kinetic friction gives:

$$F_k = \mu_k W \cos \theta$$

As θ increases, the value of cosine decreases, and both the perpendicular component of the weight and the normal force also decrease. Therefore, because kinetic friction is proportional to the normal force, **kinetic friction must decrease**.

(Choices B and D) As the angle of inclination increases, the weight component parallel to the ramp increases but does not affect kinetic friction directly.

(Choice C) Although kinetic friction increases if the weight component perpendicular to the ramp increases, this occurs when the angle of inclination decreases, not increases.

Educational objective:

The kinetic friction force between two surfaces is proportional to the normal force exerted by the surfaces. For an object on a ramp, the normal force is equal to the perpendicular component of object's weight, which decreases as the angle of inclination increases.

Time spent: 0 sec
Subject:
Physics

QID: 401019
System:
Mechanics and Energy

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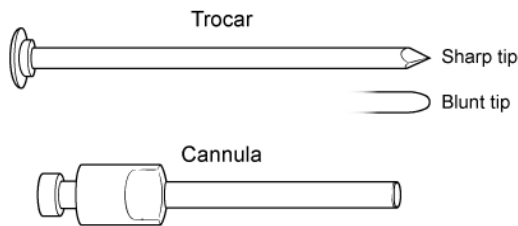


Figure 1 Cannula, trocar, and two types of tips

Low-friction trocars increase surgical precision and reduce postoperative scarring. Friction between the skin and trocar tip is tested with synthetic skin and a stationary ramp with a surface made of the same material as the trocar. For testing, the synthetic skin is pressed against the ramp's surface by a load, weighted to allow the skin to slide down at a constant speed (Figure 2). In this setup, the weight of the synthetic skin layer is negligible.

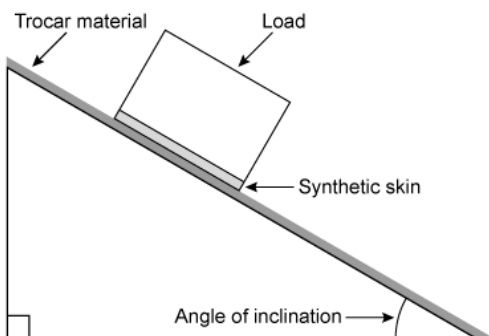


Figure 2 Experimental setup to test the friction between trocar and skin

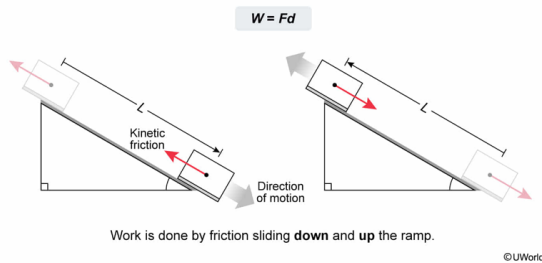
Which of the following expressions gives the magnitude of the work done by the frictional force F if the synthetic skin and the load slide down the entire length L of the ramp and then are pushed back up to their original position?

- ☐ A. 0 [37%]
- ☐ B. $(F)(L)$ [5%]
- ✓ ☒ C. $(F)(2L)$ [55%]
- ☐ D. $(F)(3L)$ [1%]

Correct

55% answered correctly

Explanation:



Friction is the force that resists sliding between two surfaces, and it is a **nonconservative force**. In the presence of a nonconservative force, the total mechanical energy E of a system (the sum of **potential energy** PE and **kinetic energy** KE) is not conserved because it is dissipated as other forms of energy, such as heat and sound. The **work** W_f done by friction equals the change in the total mechanical energy ΔE :

$$W_f = \Delta E$$

As a result, W_f depends on the **total distance** through which the force is applied rather than the displacement. As the load slides down the ramp of length L , the frictional force F acts in the opposite direction (ie, up the ramp), hence W_f equals:

$$W_{f,\text{down}} = -FL$$

When the load slides up the ramp, F again opposes the motion (ie, acts downward), thus:

$$W_{f,\text{up}} = -FL$$

The total amount of work done equals the sum:

$$W_f = W_{f,\text{down}} + W_{f,\text{up}}$$

$$W_f = (-FL) + (-FL) = -(F)(2L)$$

Therefore, the expression that gives the magnitude (ie, the absolute value) of the work done by friction is:

$$|W_f| = (F)(2L)$$

(Choice A) The work done by a conservative force (eg, gravity, electrostatic) depends on the net displacement (not distance) the force is applied. Because there is no net displacement in this case, the work done by gravity would be zero. However, friction is not a conservative force.

(Choices B and D) The total distance up and down the incline is $2L$, not L or $3L$.

Educational objective:

Friction is a nonconservative force, which does not conserve the total mechanical energy (potential plus kinetic) in a system. The magnitude of the work W done by a nonconservative force depends on the total distance d through which the force F is applied: $W = Fd$.

Time spent: 0 sec

Subject:
Physics

QID: 401020

System:
Mechanics and Energy

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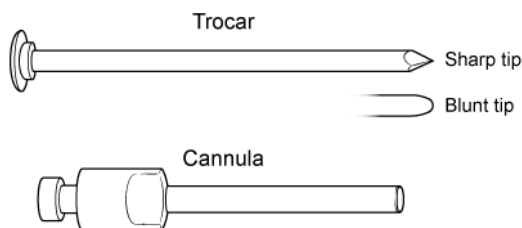


Figure 1 Cannula, trocar, and two types of tips

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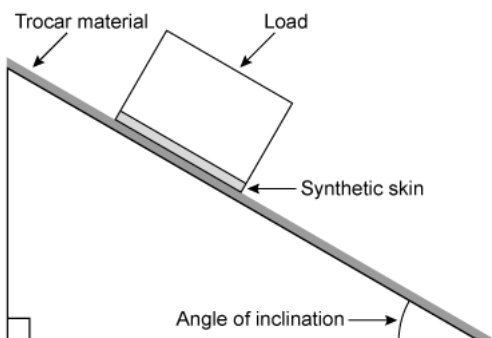


Figure 2 Experimental setup to test the friction between trocar and skin

When the load slides down the ramp during the experiment, the magnitude of the kinetic friction force between the ramp and the synthetic skin depends on:

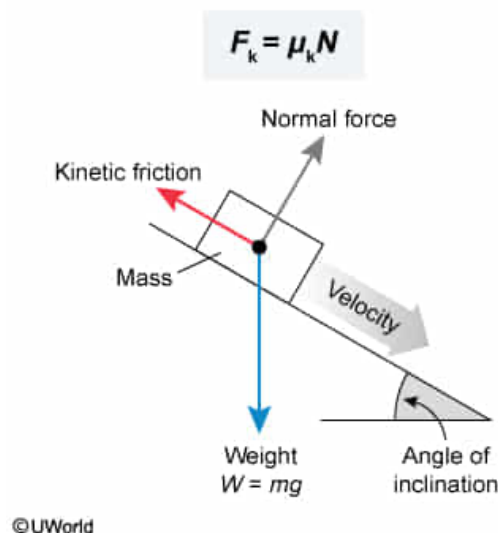
- I. the area of the contact surface.
- II. the mass of the load.
- III. the speed of the synthetic skin.

- ☐ A. I only [5%]
- ✓ ☒ B. II only [70%]
- ☐ C. I and III only [5%]
- ☐ D. II and III only [18%]

Correct

71% answered correctly

Explanation:



Kinetic friction is the frictional force between two sliding surfaces, and it is the product of the coefficient of kinetic friction μ_k (specific to the two surfaces) and the normal force N (perpendicular force a surface exerts on another):

$$F_k = \mu_k N$$

Therefore, the magnitude of the frictional force is **proportional to N** . In the case of the experiment described in the passage, the synthetic skin is resting on a ramp (Figure 2), and therefore the load's **weight determines N** . Because **weight** is proportional to mass ($W = mg$), the **mass of the load** determines the magnitude of kinetic friction in the experiment (**Number II**).

(Number I) Friction is determined by the normal force between two surfaces, which is independent of the area of the contact surface; contact areas of unequal size that have the same normal force applied experience the same amount of friction. Therefore, friction is **independent of surface area**.

(Number III) Because the value of the coefficient of kinetic friction μ_k depends only on the properties of the two surfaces, the magnitude of kinetic friction when the load slides down the ramp is **independent of speed**.

Educational objective:

The mass of an object determines its weight, which determines the normal force of an object sliding on a surface. Because kinetic friction is proportional to the normal force ($F_k = \mu_k N$), kinetic friction depends on an object's mass. The coefficient of kinetic friction μ_k is independent of the size of the contact surface or the relative speed between the surfaces.

Time spent: 0 sec

Subject:
Physics

QID: 401021

System:
Mechanics and Energy